

Alex Tanthiptham

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Summary

Experience in computer vision with OpenCV and Pytorch, React web development, PCB design. Open to relocation and travel.

Education

The University of Illinois at Urbana-Champaign

Graduated: December 2023

Bachelor of Science in Electrical and Computer Engineering, Minor in Computer Science

Course Assistantship: Data Structures (CS225), Department of Computer Science

Relevant Coursework: Digital Systems, Safe Autonomy, Data Science & Engineering, Web Programming, UI Design, Artificial Intelligence

Work Experience

Grainger College of Engineering, Cyber-Physical Systems Integration Lab (CPSIL)

August 2023 - Present

Research Assistant, VISAT Project

Urbana-Champaign, IL

- Research group aimed at improving safety and robustness of next generation autonomous vehicles under Professor Lui Sha.
- Engineered PyTorch-based ensemble networks to improve traffic sign classification accuracy of multitask learning and multi-layer perceptron networks by 7.96% and reduce loss of accuracy against ImageNet-C attacks by up to 17.83%.
- Developed robust internal tools for image pre-processing and executing adversarial attacks, utilizing PyTorch and OpenCV, to improve robustness and streamline testing.
- Applied multiprocessing to streamline experimental processes, reducing testing and processing run time by up to 65%.

Undergraduate Research Assistant, Biped Project

- Designed and assembled new IMU PCBs in KiCAD for Biped, a two-wheeled self-stabilizing research and educational robot.
- Executed comprehensive unit testing using oscilloscopes, multimeters to validate the functionality of Biped PCBs.
- Developed procedures to resolve hardware, software and mechanical faults, quadrupling the number of functional units.

Reference:

- Simon Yu, VISAT and Biped Project Lead, jundayu2@illinois.edu

FACES Consulting at UIUC

January 2023 - August 2023

Senior Consultant

Urbana-Champaign, IL

- Strategized development and launch of a desktop-based business suite for a startup's cloud collaboration app.
- Conducted customer research and interviews to identify areas of improvement, prototyped suggested features using Figma.
- Coordinated team of 4 consultants, scheduling and coordinating project progress using Gantt charts and MS Teams.

Thai Students' Association

August 2021 - December 2022

Lead Event Coordinator, Executive Board

Urbana-Champaign, IL

- Re-established the organization and revitalized its campus presence following years of inactivity due to the COVID-19 pandemic, increasing the number of active officers from 7 to 20 within a year.
- Coordinated COVID-19 relief fundraising efforts in collaboration with 16 other Thai Students Associations across the United States, raising 2,702,823 THB (approximately 89,193.16 USD) for hospitals and charities in Thailand.
- Managed a team of 5 to plan, promote, and execute community building and cultural outreach events in both the United States and Thailand for over 80 Thai-American and international student attendees from up to 5 different universities.

Selected Projects

TagAlong Personal Assistant Robot tinyurl.com/5e5js8ee

January 2023 - May 2023

- Developed hardware and software for a user-following cargo carrying robot using a microcontroller for data processing and PID motor control with RaspberryPi for OpenCV tracking of the user in Cython.
- Designed modular main and breakout board PCBs in KiCAD, assembling boards using both hand and reflow soldering.
- Engineered power subsystem for 3.3, 5 and 12V outputs from SLA battery using Buck converters and linear regulators.
- Awarded an Honorable Mention award for Spring 2023 Senior Design Project.

Resume Review Web App

October 2022 - December 2022

- Developed a MERN (MongoDB, Express JS, React, Node JS) stack web app that allows users to upload their resumes, tag them by fields and receive as well as give anonymous reviews from other users and experts.
- Engineered a MongoDB backend and RESTful API using Express, Node JS and Vercel to parse, store, retrieve and display PDF files via Multer, and integrated with Firebase for user authentication and user data.
- Designed an interactive UI/UX prototype using Figma to gauge user feedback and iterate on design.

FPGA Keyboard Synthesizer

October 2021 - December 2021

- Designed a USB keyboard controlled PCM audio storage and playback system for the Terasic DE10-Lite FPGA.
- Implemented a driver for USB keyboards using C on embedded NIOS II processors and PCM digital audio processing using C and SystemVerilog, verifying circuit functionality using Quartus SignalTap logic analyzer and RTL viewer.
- Documented detailed technical specifications including FPGA resource usage, RTL diagrams and module functions.

Skills: LTSpice, KiCad, Scopy, Quartus Prime, Fritzing CAD, System Verilog, C, C++, Python, Pytorch, SciPy, Scikit-Learn, Keras, TensorFlow, OpenCV, Numpy, Java, Javascript, React, Typescript, Express JS, NodeJS, Mongoose JS, Android Studio, MongoDB, Postman, HTML, CSS, Jupyter Notebook, Docker, Soldering, Thai (Native), English (Native)